Царенов Олег ПИ 22/2

Задание 3

1. Подключился к серверу по SSH

```
imac@MacBook-Air-imac ~ % ssh u82290@kubsu-dev.ru -p 58528
u82290@kubsu-dev.ru's password:
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Apr 23 06:17:17 PM UTC 2026

  System load:  0.0                Processes:               210
  Usage of /:   25.9% of 48.91GB   Users logged in:         4
  Memory usage: 11%                IPv4 address for enp6s18: 192.168.199.8
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

83 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Fri Apr 10 05:41:30 2026 from 10.16.4.131
u82290@web-server:~$
```

2. Зашел в MySQL

```
u82290@web-server:~$ mysql -u u82290 -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 638
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

3. Выбрал свою базу данных

```
mysql> USE u82290;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql>
```

4. Посмотрел все заявки

увидел таблицу с колонками: id, full_name, phone, email, birth_date, gender, bio, contract_agreed, created_at.

```
mysql> SELECT * FROM applications;
+----+-------------------------+------------------+----------------------------+------------+
| id | full_name               | phone            | email                      | birth_date |
+----+-------------------------+------------------+----------------------------+------------+
|  1 | Царенов Олег            | +7 932 404 82 04 | mac.tsarenov2006@gmail.com | 2006-07-08 |
|  2 | Царенов Олег            | +7 932 404 82 04 | mac.tsarenov2006@gmail.com | 2006-07-08 |
|  3 | Царенов Олег            | +7 932 404 82 04 | mac.tsarenov2006@gmail.com | 2006-07-08 |
|  4 | Царенов Олег            | +7 932 404 82 04 | mac.tsarenov2006@gmail.com | 2006-07-08 |
|  5 | Царенов Олег            | +7 932 404 82 04 | mac.tsarenov2006@gmail.com | 2002-03-04 |
|  6 | Царенов Олег            | +7 932 404 82 04 | mac.tsarenov2006@gmail.com | 2001-03-09 |
|  7 | ол                      | 79999999999      | mac.tsarenov2006@gmail.com | 2026-01-01 |
|  8 | Царенов Олег            | +7 932 404 82 04 | mac.tsarenov2006@gmail.com | 2001-07-08 |
+----+-------------------------+------------------+----------------------------+------------+
8 rows in set (0.01 sec)

mysql>
```

```
--------+-------------------------------------------------+-----------------+---------------------+
 gender | bio                                             | contract_agreed | created_at          |
--------+-------------------------------------------------+-----------------+---------------------+
 male   | я Царенов Олег                                  |               1 | 2026-04-09 19:52:48 |
 male   | я Царенов Олег                                  |               1 | 2026-04-09 20:02:58 |
 male   | я Царенов Олег                                  |               1 | 2026-04-09 20:03:24 |
 male   | hello                                           |               1 | 2026-04-09 20:08:32 |
 male   | да это снова я Царенов Олег                     |               1 | 2026-04-09 20:21:44 |
 male   | ыыы                                             |               1 | 2026-04-09 20:35:19 |
 male   | проверка1                                       |               1 | 2026-04-09 20:57:43 |
 male   | это снова я                                     |               1 | 2026-04-23 18:11:48 |
--------+-------------------------------------------------+-----------------+---------------------+
```

5. Посмотрел выбранные языки для конкретной заявки номер 1

```
mysql> SELECT p.name
    -> FROM application_languages al
    -> JOIN programming_languages p ON al.language_id = p.id
    -> WHERE al.application_id = 1;
+--------+
| name   |
+--------+
| Pascal |
+--------+
1 row in set (0.01 sec)

mysql>
```

6. Удобный сводный запрос (все заявки с языками)

```
mysql> SELECT
    ->     a.id,
    ->     a.full_name,
    ->     a.email,
    ->     GROUP_CONCAT(p.name SEPARATOR ', ') AS languages
    -> FROM applications a
    -> LEFT JOIN application_languages al ON a.id = al.application_id
    -> LEFT JOIN programming_languages p ON al.language_id = p.id
    -> GROUP BY a.id;
+----+-----------------------+-------------------------+-----------------------------------------------------------------------------------------+
| id | full_name             | email                   | languages                                                                               |
+----+-----------------------+-------------------------+-----------------------------------------------------------------------------------------+
|  1 | Царенов Олег          | mac.tsarenov2006@gmail.com | Pascal                                                                                  |
|  2 | Царенов Олег          | mac.tsarenov2006@gmail.com | Pascal                                                                                  |
|  3 | Царенов Олег          | mac.tsarenov2006@gmail.com | Pascal                                                                                  |
|  4 | Царенов Олег          | mac.tsarenov2006@gmail.com | Pascal                                                                                  |
|  5 | Царенов Олег          | mac.tsarenov2006@gmail.com | JavaScript                                                                              |
|  6 | Царенов Олег          | mac.tsarenov2006@gmail.com | Python                                                                                  |
|  7 | ол                    | mac.tsarenov2006@gmail.com | Pascal, C, C++, JavaScript, PHP, Python, Java, Haskell, Clojure, Prolog, Scala, Go |
|  8 | Царенов Олег          | mac.tsarenov2006@gmail.com | JavaScript                                                                              |
+----+-----------------------+-------------------------+-----------------------------------------------------------------------------------------+
8 rows in set (0.00 sec)
```

7. Проверяем, таблицы и их структуру

```
mysql> SHOW TABLES;
+------------------------+
| Tables_in_u82290       |
+------------------------+
| application_languages  |
| applications           |
| programming_languages  |
+------------------------+
3 rows in set (0.00 sec)

mysql> DESCRIBE applications;
+-----------------+-----------------------+------+-----+--------------------+-------------------+
| Field           | Type                  | Null | Key | Default            | Extra             |
+-----------------+-----------------------+------+-----+--------------------+-------------------+
| id              | int unsigned          | NO   | PRI | NULL               | auto_increment    |
| full_name       | varchar(150)          | NO   |     | NULL               |                   |
| phone           | varchar(20)           | NO   |     | NULL               |                   |
| email           | varchar(100)          | NO   |     | NULL               |                   |
| birth_date      | date                  | NO   |     | NULL               |                   |
| gender          | enum('male','female') | NO   |     | NULL               |                   |
| bio             | text                  | NO   |     | NULL               |                   |
| contract_agreed | tinyint(1)            | NO   |     | 0                  |                   |
| created_at      | timestamp             | YES  |     | CURRENT_TIMESTAMP  | DEFAULT_GENERATED |
+-----------------+-----------------------+------+-----+--------------------+-------------------+
9 rows in set (0.01 sec)
```

applications – сущность заявки.
programming_languages – справочник языков.
application_languages – связь «многие ко многим»

Каждая неключевая колонка должна зависеть только от первичного ключа всей таблицы, а не от части ключа или другой неключевой колонки. Если бы мы хранили языки прямо в заявке, например, в одном поле через запятую ("PHP, JavaScript") или в нескольких столбцах (language1, language2…), это нарушало бы первую нормальную форму и создавало бы избыточность. Для построения отчётов (например, «сколько заявок выбрали Python») пришлось бы разбирать строки, что медленно и неудобно.

8. Вышел из MySQL

```
mysql> EXIT;
Bye
u82290@web-server:~$
```